HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Led client discovery sessions and site visits to translate operational problems into actionable requirements and defined technical scopes for **manufacturing scheduling** solutions.
- Developed optimisation algorithms for **production planning**, incorporating complex constraints such as routings, setups, and resource availability to improve scheduling efficiency by **18%**.
- Analysed large-scale **production data** from client operations, assessing data quality and deriving key model inputs for predictive maintenance tools, identifying anomalies that led to a **15%** reduction in unplanned downtime.
- Designed and implemented **Python** scripts for automated data extraction and transformation from diverse sources, building ETL pipelines that reduced manual data preparation time by **30 hours** per week.
- Collaborated with software engineering teams to integrate analytical tools and models into production systems, ensuring seamless deployment and operational handover for client-facing applications.
- Contributed technical content to multiple **European research proposals**, including effort estimates and validation plans, supporting successful bids valued at over **€1.2M**.
- Managed CI/CD pipelines using **Jenkins** and **Git** for automated testing and deployment of data analytics solutions within an **AWS** cloud environment.

SKILLS

Programming Languages – Python, SQL, Java, R

Data Analytics & ML – Power BI, Tableau, scikit-learn, pandas, NumPy, DAX, Machine Learning, Statistical Analysis, Optimisation, Scheduling

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Databases – PostgreSQL, MySQL, Oracle

Tools & Methodologies – Jira, Confluence, Agile, Scrum, REST APIs, Linux Shell Scripting

Domain Expertise – Manufacturing Operations, Production Planning, Scheduling, Data Quality Assessment, Client Engagements, Research Proposals

PROJECTS

Client Production Scheduling Optimisation Tool

- Developed a **Python**-based scheduling engine that modelled complex manufacturing constraints, including machine availability, shift patterns, and material flow, for a key industrial client.
- Integrated the scheduling engine with client's existing ERP system via **REST APIs**, enabling dynamic updates and real-time adjustments to production plans, improving on-time delivery rates by **12%**.
- Designed and delivered technical documentation and training materials for the client's planning team, ensuring successful adoption and operational use of the new tool.

Predictive Maintenance Analytics for Manufacturing

- Engineered **data pipelines** using Python and SQL to ingest sensor and operational data from industrial equipment, aggregating over **500,000** data points daily for analysis.
- Applied **machine learning** techniques (scikit-learn) to build predictive models that identified failure patterns, achieving an accuracy of **88%** in forecasting equipment maintenance needs.
- Developed interactive **Power BI dashboards** to visualise equipment health, predicted failure risks, and recommended proactive maintenance schedules for operational managers.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)